

Contents

Abstract	2
Introduction	3
Literature Review	5
Value-at-Risk Methodology	5
Historic Simulations	6
Volatility Models	7
Covariance Var	8
Monte Carlo Simulations	9
LSTM	10
Comparative Studies	12
Research Gaps	14
Research Objectives	16
Data	17
Portfolio Creation	23
Data Analysis	25
Results & Interpretation	30
Conclusion and Further Studies	32
References	37

Abstract

This study presents a comparative analysis of Value at Risk (VaR) estimation methods, encompassing traditional, statistical, and artificial intelligence-based approaches. By evaluating models such as Historical VaR, Covariance VaR, Monte Carlo simulation, and Long Short-Term Memory (LSTM) neural networks, we assess their predictive accuracy, computational efficiency, and robustness under varying market conditions. The research employs backtesting techniques to validate the reliability of each method, revealing that AI-driven models, particularly LSTM, outperform conventional techniques in capturing volatility patterns and extreme losses. Despite their superior predictive performance, AI-based models pose challenges related to interpretability and real-world implementation. This paper identifies gaps in existing literature, emphasizing the need for hybrid models that balance accuracy and explainability. Furthermore, we propose future research directions, including the integration of generative models for risk prediction, the development of interpretable deep learning frameworks, and stress-testing AI-driven VaR models under extreme market conditions. The findings contribute to the ongoing discourse on risk management in finance, underscoring the potential of deep learning in enhancing financial stability and decision-making.

Introduction

Due to increased volatility and unpredictability in global financial markets, as well as the massive development in trading activity and foreign exposure, there is a growing need for trustworthy risk management strategies. One-need-only

Consider the losses from previous stock market and currency crashes, as well as those brought on by the risky positions taken, for instance, by Metallgesellschaft, Barings, Daiwa, and Orange County. One such risk management strategy created to enhance the control of downside risk is value-at-risk, or VaR. By calculating the worst expected loss over a selected time horizon within a specified confidence interval, it seeks to summarise risk. Because the value-at-risk concept is predicated on the likelihood of significant negative returns, it emphasises how financial institutions have had to become increasingly focused on controlling this downside risk. Only by employing these risk management techniques can the exposure to significant negative fluctuations in financial markets be managed and minimised. Their advantage, meanwhile, mainly depends on how accurate the value-at-risk estimations are.

The probability distribution of the anticipated returns yields the VaR estimate. This suggests that one must make assumptions about the true shape of the anticipated return distribution. This can be done by assuming that the returns are taken from a certain statistical distribution or by assuming that the distribution of the predicted returns equals the empirical distribution based on historical observations. The precise structure of these analytical distributions depends on a number of parameters that are calculated from historical data and that have more recently made it possible to employ conditioning techniques like generalised autoregressive conditionally heteroskedastic (GARCH) processes.

Since it allows for easy conversion (between quantiles and temporal horizons) and is therefore more practical within the Basle Committee's framework, a parametric approach has been the recommended solution. Additionally, it makes it simple to add conditionality in the data into the VaR estimate, which improves the suitability of VaR forecasts. Therefore, it is essential to assume that the specified distribution accurately represents the distribution in

the left tail, which represents the negative returns, in order to estimate the VaR.

VaR estimate can contain significant errors if there is a difference between the empirical distribution and the parametric distribution.

Asset returns are frequently taken to be normally or lognormally distributed for ease and simplicity. It has been demonstrated, nevertheless, that the return distributions on a large number of assets have broader tails than the normal distribution. One This indicates that when one moves farther into the tails, the VaR is underestimated due to the presumption of normalcy. When attempting to determine an asset's VaR, the precise nature of this additional mass in the distribution's tails is essential. It has previously been challenging to apply other fatter-tailed distributions, such as Pareto and sum-stable distributions. Thus, the most practical stand-in for the real distribution of an asset has remained the normal distribution. Nonetheless, the wider the deviation from the normal distribution and the greater the mistakes in VaR estimation, the fatter the tails of the asset return distribution under consideration. The million-dollar positions that mutual funds, for instance, usually hold magnify these inaccuracies.

Therefore, straightforward methods for estimating VaR that account for the tail fatness visible in return distributions are required. We describe such a basic method in this paper. We demonstrate that for distributions showing a larger probability of substantial negative returns, VaR-x estimates—VaR estimates derived from assuming the Student t-distribution as at to the empirical distribution—are better equipped to represent the additional risk involved. We are interested in the negative returns connected to the left tail of the distribution since we are examining downside risk. The amount of fatness in the tail of the return's distribution is directly measured by the Student t-distribution's tail parameter, which is reflected by its number of degrees of freedom and set equal to the tail index for the left tail. Compared to the normal distribution, this approach has numerous benefits. Fat tails are first caught. Second, we can avoid assuming that distributions are symmetric by concentrating solely on the left tail. There is a chance that the distribution may be skewed because the tail fatness

may differ between the two tails of the return distribution. In contrast to what would be achieved by assuming normalcy, this gives us a straightforward and more accurate estimator.

Literature Review

Value-at-Risk Methodology

An estimate of the VaR can be used to condense exposure to downside risk into a single figure. "The worst expected loss over a target horizon within a given confidence level" is how Jorion (1996) formally defines this. In accordance with Jorion, we define R as the anticipated return over the goal horizon and W_0 as the original investment. W is defined as the lowest value of the portfolio at the specified confidence level c ; in other words, the portfolio value should not, with probability c , fall below W . VaR is defined as the dollar loss in relation to the portfolio's projected mean value.

$$VaR = E(W) - W^* \quad (1)$$

Defining R as the expected return associated with the portfolio value W ,

$$W^* = W_0(1 + R^*) \quad (2)$$

gives us the VaR measured as the dollar loss relative to the mean,

$$VaR = -W_0(R^* - \mu) \quad (3)$$

where μ is the expected return on the portfolio for the target horizon. Estimating the cutoff return R is crucial to being able to deliver an accurate VaR estimate. Since the cutoff return calculated from historical data is the main focus of this work, we presume that it is representative of the projected return measure.

There are two categories of statistical techniques used to best estimate these cutoff returns: parametric and nonparametric. The simplest straightforward nonparametric method calcu-

lates an estimate by using the historical distribution itself. Direct empirical estimation of the VaR. In the parametric scenario, a statistical distribution whose characteristic parameters are taken from the historical data is used to try to fit the historical distribution. Therefore, before introducing the mechanism underlying VaR-x, we will briefly discuss the common methods for estimating VaR. The primary distinction between VaR and VaR-x is that the latter improves the quality of the estimates in a straightforward and effective manner by incorporating the tail fatness evident in financial returns into the VaR estimate.

The following integral for the distribution of expected returns $f(r)$ yields the cutoff return, which is defined as the worst possible realisation R with a confidence level c :

$$1 - c = \int_{-\infty}^{R^*} f(r) dr \quad (4)$$

Historic Simulations

The most popular non-parametric method is historical simulation. $\text{VaR}(\alpha)$ is the, quantile of the empirical distribution since this method approximates $F(r)$ using the empirical distribution of financial returns. Various sample sizes can be taken into consideration when calculating the empirical distribution of financial returns. Down (2002) has thoroughly documented the Historical Simulation's benefits and drawbacks. The following are the two primary advantages: The method is fairly simple to apply, and it can handle wide tails, skewness, and other non-normal aspects in financial observations since it does not rely on parametric assumptions about the return portfolio's distribution. This method's worst possible flaw is that its outcomes are entirely reliant on the data set. Historical Simulation frequently underestimates danger if our data era is abnormally quiet, and overestimates risk if our data period is abnormally turbulent. Furthermore, historical simulation techniques can occasionally be slow to capture significant events, including the spikes in risk brought on by abrupt market volatility. The Historical Simulation performed at least as well as the early approaches, the Parametric approach and the Monte Carlo simulation, according to the

first papers comparing VaR methodologies, including those by Beder (1995, 1996), Hendricks (1996), and Prittsker (1997).

Volatility Models

The GARCH family, stochastic volatility models, and realised volatility-based models are the three categories into which the volatility models put out in the literature to describe the features of financial returns can be separated. Regarding the GARCH family, the Autoregressive Conditional Heteroscedasticity (ARCH), which was proposed by Engle (1982), has a variance that fluctuates over time rather than staying constant. By adding the ARCH generalised model (GARCH), Bollerslev (1986) expanded the model even more. Two equations are specified and estimated using this model: the first shows how returns have changed over time in relation to previous returns, while the second patterns how return volatility has changed over time. The GARCH (p,q) model, which is the most generalised formulation of the GARCH models, is expressed as follows:

$$r_t = \mu_t + \varepsilon_t \quad (5)$$

$$\sigma_t^2 = \alpha_0 + \sum_{i=0}^q \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2 \quad (6)$$

In the GARCH (1,1) model, the empirical applications conducted on financial series detect that $\alpha_1 + \beta_1$ is observed to be very close to the unit. Then, using the requirement that the addition in equation equals the unit, the integrated GARCH model (IGARCH) of Engle and Bollerslev (1986) is constructed. The relatively long phasing out of the shock impact on the conditional variance (volatility persistence) makes the IGARCH model's conditional variance features unappealing from an empirical standpoint. The fractional integrated GARCH model (FIGARCH), which was proposed by Baillie et al. (1996), exhibits exponential behaviour in the impacts that fade away. The simplest specification, FIGARCH (1, d, 0), is as follows:

$$\sigma_t^2 = \frac{\alpha_0}{1 - \beta_1} + \left(1 - \frac{(1 - L)^d}{1 - \beta_1 L}\right) r_t^2 \quad (7)$$

The conditional variance of the model is probably positive for all t cases if the parameters meet the setup constraints $\alpha_0 > 0$, $0 < d \leq 1$. The r_t^2 effect on σ_{t+k}^2 is likely to cause a decrease over the hyperbolic rate as k increases in this scenario. Although the aforementioned models accurately characterise the volatility clustering properties, they do not account for the asymmetric performance of yields prior to positive or negative shocks (leverage effect), which means that they do not fully capture the nature of the volatility of the financial times series. The effect of positive innovations is the same as the effect of negative innovations of equal absolute value since earlier models rely on square mistakes. However, empirical evidence indicates that the leverage effect is present in financial time series, meaning that volatility rises more quickly when rates are negative than when they are positive. Several non-linear GARCH formulations have been proposed to represent the leverage effect. Table 1 lists a few of the most well-liked. Bollerslev (2009) provides a thorough analysis of the asymmetric GARCH models.

Covariance Var

The variance-covariance approach was initially popularised by J.P. Morgan's RiskMetrics framework (J.P. Morgan, 1996) and is based on the notion that asset returns are normally distributed. This assumption enables the calculation of VaR using statistical features of returns, particularly standard deviation (volatility) and correlation with other assets in a portfolio. Studies show that the covariance-based approach provides a reasonable estimate of potential losses under normal market conditions (Hull, 2015).

The dependence of covariance-based VaR on the normalcy assumption is one of the main criticisms levelled at it. According to empirical research (e.g., Cont, 2001), financial returns frequently show skewness and fat tails, which understates extreme risks. Furthermore, correlation breakdown—a process whereby correlations between assets tend to increase—occurs

during financial crises (Longin Solnik, 2001). Because of this, when covariance-based VaR estimates are most needed, they may be deceptive.

Researchers have suggested a number of changes to covariance-based VaR in order to overcome its drawbacks. In order to improve risk estimations during tumultuous times, Engle (2002) created Generalised Autoregressive Conditional Heteroskedasticity (GARCH) models, which account for time-varying volatility. Copula Models (Embrechts et al., 2003) are another enhancement; they model dependencies more flexibly, which relaxes the condition of normalcy. In order to improve predicted accuracy, hybrid models that incorporate machine learning approaches have also been investigated (Huang et al., 2020).

The effectiveness of covariance-based VaR in actual financial markets has been evaluated in a number of empirical investigations. In their research of large portfolios, Pafka Kondor (2001) discovered that although the covariance-based approach works well when markets are calm, it drastically understates risk during times of crisis. Likewise, a comparison study by Danielsson (2011) showed that Monte Carlo and historical simulation techniques frequently yield more precise risk assessments in non-normal settings. Nonetheless, covariance-based VaR is still often utilised in banking and regulatory procedures because of its simplicity and convenience of usage (Basel Committee, 2019).

Monte Carlo Simulations

Simulating N draws from the distribution of returns on date $t + 1$ is the most straightforward Monte Carlo method for estimating the VaR on date t on a one-day horizon at a 99 significance level. After sorting the N distinct draws from the one-day returns, the VaR estimate is empirically determined as the VaR quantile of the simulated distribution of returns. The VaR at a 99 level is calculated by reading off element $N/100$. The key non-normal characteristics of their multi-variate conditional distributions and path-dependent behaviour, like volatility clustering, must be captured by applying simulations to a dynamic model of risk factor returns. Regarding the first of these, the tendency of high-frequency returns to exhibit clustered volatility is one of their most significant characteristics. Here, we may use

the simulation returns from the previous simulation to get the GARCH variance estimate at time t . We can then put $r_t = \hat{\sigma}_t z_t$, where z_t is a simulation from a standard normal variable. Regarding the second item, we can use copulas as the dependent metric in place of correlation, or we can use the standard multivariate normal or t-Student distribution to model the interdependence. Monte Carlo is an interesting technique that may be used to estimate the VaR for non-linear portfolios (see Estrella et al., 1994) because it requires no simplifying assumptions regarding the joint distribution of the underlying data. Nevertheless, it entails significant computational costs. Its application to actual risk containment issues has been limited by this expense. Alternative techniques with the lowest computing costs were suggested by Srinivasan and Shah (2001), and Antonelli while Iovino (2002) put out a methodology that increases the computational efficiency of the Monte Carlo simulation approach to VaR estimates. Lastly, research comparing VaR approaches has demonstrated that methods other than Monte Carlo produce more accurate VaR estimates (see Abad and Benito, 2013; Huang, 2009; Tolikas et al., 2007; Bao et al., 2006). In conclusion, we have examined some of the most significant VaR methodologies in this part, ranging from the conventional models to the more modern techniques. Theoretically, there are benefits and drawbacks to each of these strategies. We restate these benefits and drawbacks in Table 3. We shall examine the outcomes of these approaches from a practical standpoint in the upcoming sections.

LSTM

The variance-covariance approach is predicated on the idea that asset returns are normally distributed, and it was first made famous by J.P. Morgan's RiskMetrics framework (J.P. Morgan, 1996). This presumption makes it possible to calculate VaR utilising statistical characteristics of returns, specifically their correlation with other assets in a portfolio and standard deviation (volatility). According to studies, the covariance-based technique offers a fair assessment of possible losses in typical market circumstances (Hull, 2015).

In multi-asset portfolios, where interdependencies among several assets are critical to over-

all risk exposure, covariance-based VaR is particularly helpful. Alexander's (2008) research demonstrates how this strategy enables businesses to create an effective frontier while maximising risk-adjusted earnings. However, precise estimation of the covariance matrix—which is susceptible to changes in market volatility—is necessary for the strategy to be effective.

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Comparative Studies

Estimation of Value at Risk (VaR) has undergone a remarkable transformation over the years, from classical approaches to statistical models and more recently, artificial intelligence (AI)-driven methods. VaR is a popular risk measure in financial markets, measuring the maximum possible loss of a portfolio within a specific time horizon at a given confidence level. Several studies have investigated comparative performance of these methods, including accuracy, computational efficacy, and generalizability across a range of market conditions.

Old VaR models, including the Historical Simulation and Covariance (Parametric) Method, are risk management staples due to ease of interpretation and simplicity. The Historical Simulation method uses historical return distributions to proxy potential losses and is therefore non-parametric and distribution-independent. Nonetheless, it has been found that it tends to underestimate risk in turbulent or crisis times because it does not reflect on unprecedented market occurrences. The Covariance Method, by contrast, presumes a normal return distribution, utilizing mean and standard deviation to determine VaR. Although computationally efficient, it has difficulty replicating the fat-tailedness of financial returns, resulting in inaccuracies in times of high market volatility.

In order to overcome these limitations, more sophisticated statistical methods, including Monte Carlo Simulation, have been thoroughly researched. Monte Carlo methods create many hypothetical scenarios of returns from a specified probability distribution, giving an increased degree of flexibility to model complicated risk structures. Research has identified that Monte Carlo Simulation offers improved risk estimates compared to conventional methods, particularly when including non-normal distributions. But its performance is highly sensitive to the assumptions about the underlying distribution, and it can be computationally expensive when modeling large datasets or complex dependencies among financial assets. Recent developments in AI-driven VaR models, especially those based on machine learning and deep learning methods, have shown considerable promise in enhancing risk estimation. Long Short-Term Memory (LSTM) neural networks, a kind of recurrent neural network

(RNN), have become popular for capturing the non-linear relationships and volatility clustering in financial time series data. Empirical evidence has demonstrated that LSTM-based VaR models perform better than conventional methods by adapting dynamically to evolving market conditions and identifying risk patterns that statistical models cannot capture. Nevertheless, although AI-driven models have increased predictive accuracy, they incur a loss in interpretability. This accuracy versus transparency trade-off has been an essential issue in finance research since regulatory authorities and risk managers usually desire models with good explanations of predictions.

Comparative research comparing VaR estimates from various models indicates that although AI-based methods offer better predictive accuracy, they consume high computational power and large amounts of data for training. Under regular market conditions, conventional and statistical approaches can yield fairly accurate VaR estimates with less computational effort. But during financial crises or periods of high volatility, AI-based models clearly outperform by better capturing sudden changes in the market. Other research also points to the necessity of hybrid methods that merge statistical and AI-based models in order to reconcile interpretability and accuracy.

Results from backtesting across different research studies suggest that AI-based models have lower failure rates in VaR estimation, i.e., fewer occurrences of actual losses greater than estimated risk levels. But AI models may be hyperparameter-sensitive and data preprocessing method-sensitive, requiring diligent tuning and verification. A few researchers contend that although AI improves predictive power, excessive dependence on deep learning models without a domain expert's knowledge can make risk estimations unreliable, particularly where past data inadequately capture future uncertainty.

Research comparing VaR under varying market conditions confirms the notion that there is no one best method. Conventional approaches are best suited for steady conditions, statistical models offer greater flexibility, and AI-based methods are best suited for dealing with intricate, unstable environments. The future prospects for AI-based risk estimation

include explainable AI (XAI) methods, which seek to increase the transparency of deep learning models without compromising predictive capability. Moreover, combining real-time market data and alternative datasets, including news sentiment and macroeconomic data, might further improve AI-based VaR models for stronger risk assessment.

In summary, the comparative analysis of conventional, statistical, and AI-driven VaR models presents unique strengths and limitations for each method. Although AI models demonstrate impressive developments in financial risk assessment, issues with interpretability, computational power, and regulatory adherence persist. There is a need for future research to create hybrid models that utilize the most desirable attributes of various models while maintaining accuracy and usability in operational risk management scenarios.

Research Gaps

The comparative study of historical, statistical, and AI-driven Value at Risk (VaR) models in literature reveals major findings and shortcomings that define the financial risk measurement landscape today. The studies have invariably shown that although traditional approaches like Historical Simulation and Parametric methods are simple and easy to interpret, they are not able to capture extreme market movements and non-linear relationships. Sophisticated statistical models such as Monte Carlo Simulation offer more flexibility at the expense of requiring sensible distributional assumptions, thus being less effective during times of financial stress. AI-based VaR models, especially those based on deep learning architectures such as Long Short-Term Memory (LSTM) networks, have proved to possess better forecasting capabilities by adjusting dynamically to the fluctuations in the market. However, despite their strong performance, these models remain largely experimental, with limited real-world implementation in financial institutions due to challenges in explainability and regulatory acceptance.

A major gap identified in the literature is the lack of practical application of AI-based VaR models in live trading and risk management environments. Although various research has established that machine learning models have high performances when applied in back-

tested environments, they are still not widely adopted in financial institutions due to issues of transparency, data integrity, and high computational demands. Regulators and risk managers need models that yield transparent explanations for risk estimations, a component that most AI models have yet to develop. This calls for hybrid methods that incorporate conventional financial theories with deep learning abilities to facilitate more interpretable and accurate risk predictions. Statistical methods combined with AI in hybrid models may help bridge the accuracy-explainability gap, so that risk evaluation methods are still robust but interpretable.

Another aspect that is not yet fully explored is the use of AI-based VaR models for particular asset classes, like cryptocurrencies, private equity, or alternative investments. The majority of research targets stock market indices or fixed-income portfolios, so there is scope for more work to be done on how deep learning models fare in predicting risk in a variety of financial instruments. Due to the specific nature of various asset classes, AI models might need to be tailored with special architectures and training data to provide sound risk predictions. The paucity of high-quality, asset-specific datasets also makes it challenging to develop strong AI-based risk assessment models.

Future research should explore the potential of generative models, such as Generative Adversarial Networks (GANs), in financial risk prediction. GANs have been successfully used in various domains to generate synthetic data that mimic real-world distributions, which could be particularly useful for stress testing VaR models under extreme market conditions. By simulating financial crises or black swan events, GANs could enhance the robustness of AI-based risk models and improve their ability to predict tail-risk events.

Also, the development of explainable AI (XAI) techniques in risk management is an important area for future research. Increasing the transparency of deep learning models through feature attribution, visualization methods, or rule-based explanations can enhance their adoption in financial risk management. Researchers need to develop AI models that not only have high predictive accuracy but also offer simple understandability of the drivers be-

hind the risk estimates. This would ease regulatory compliance and enhance confidence in AI-based risk assessment approaches.

Lastly, comprehensive testing of AI-powered VaR models under stressful market conditions is an area with limited exploration. The majority of research tests model performance on past data, which might not best reflect the dynamic nature of financial markets. Research in the future should focus more on stress testing AI-based models using synthetic or actual crisis datasets to test their robustness under highly volatile scenarios. Such initiatives would make AI-based VaR models not only efficient in normal market conditions but also resilient enough to cope with financial crises.

In summary, although AI-based VaR models have shown considerable progress in risk forecasting, their practical applicability is still limited because of interpretability issues, computational complexity, and regulatory issues. There is a need for hybrid solutions blending conventional and AI-driven methods and for new approaches like GANs and explainable AI for greater transparency and resilience. Resolving these knowledge gaps will be essential to the future of risk management, ensuring that AI-driven models become realistic tools that financial institutions can comfortably implement.

Research Objectives

The primary objective of this study is to **compare and evaluate the accuracy and effectiveness of different Value at Risk (VaR) estimation methods** applied to a portfolio of Indian bank stocks. To achieve this, the study focuses on the following specific objectives:

1. **Compare Traditional, Statistical, and AI-Based VaR Models** – Assess and analyze the performance of Historical Simulation, Covariance (Parametric) Method, Monte Carlo Simulation, and LSTM-based Deep Learning in estimating portfolio risk.
2. **Evaluate the Accuracy of Each Model** – Conduct a comparative analysis between predicted VaR values and actual worst-case losses to determine which method provides

the most reliable risk estimates.

3. **Assess Model Performance in Different Market Conditions** – Investigate how different VaR models perform during stable market periods versus crisis periods, identifying their strengths and weaknesses in handling extreme market movements.
4. **Analyze Computational Efficiency and Practical Applicability** – Examine the computational complexity and feasibility of implementing each VaR model in real-world financial risk management settings.
5. **Explore the Potential of AI-Based VaR Estimation** – Assess the advantages of deep learning approaches, particularly LSTM, in capturing non-linear dependencies and volatility clustering compared to traditional VaR models.
6. **Identify Gaps and Suggest Future Enhancements** – Highlight limitations in existing methodologies and propose potential improvements, such as hybrid models that combine AI and traditional statistical approaches for better interpretability and accuracy.

These objectives aim to contribute to the ongoing research in financial risk management by determining the most effective approach for VaR estimation, with a particular emphasis on AI-driven solutions.

Data

The dataset used in this study consists of the daily stock returns of a portfolio composed of five major Indian banks: HDFC Bank, ICICI Bank, Kotak Mahindra Bank, Axis Bank, and the State Bank of India (SBI). The data spans from January 2015 to December 2024, covering a nearly decade-long period to ensure robustness in risk estimation. Stock price data has been sourced from reliable financial databases such as the National Stock Exchange (NSE) and Bombay Stock Exchange (BSE).

The dataset includes the following attributes:

- **Date:** The trading day identifier
- **Closing Price:** The adjusted closing price of each stock.
- **Daily Returns:** Computed as the logarithmic difference of consecutive closing prices.
- **Portfolio Returns:** The weighted average return of the five stocks, assuming an equal-weighted portfolio.

Before applying LSTM models, standard data preprocessing steps were conducted:

1. **Handling Missing Data:** Any missing values were imputed using forward fill or interpolation techniques.
2. **Normalization:** Stock returns were scaled using MinMax scaling to improve model convergence.
3. **Stationarity Checks:** Augmented Dickey-Fuller (ADF) tests were performed to ensure stationarity.
4. **Feature Engineering:** Volatility indicators such as rolling standard deviations and exponential moving averages were included as additional input.

This dataset provides a robust foundation for assessing the performance of LSTM-based VaR models compared to traditional approaches.

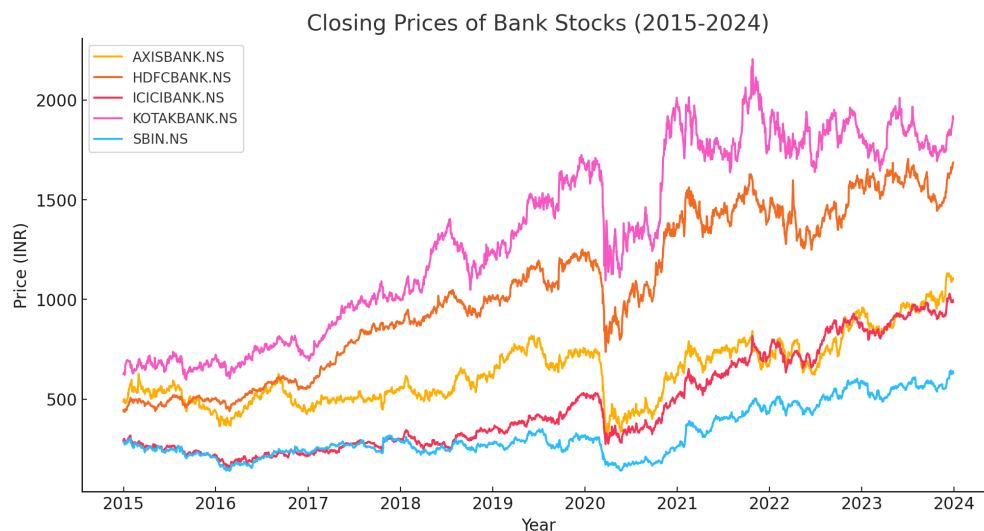


Figure 1
Closing Price Trends.

The closing price trends of the five bank stocks over the years 1 reflect their long-term performance and reaction to market conditions. The price movements exhibit significant upward trends, particularly for HDFC and ICICI, which have consistently grown over the years. Kotak Bank also follows a steady growth trajectory, while SBI and Axis Bank show more pronounced fluctuations.

The impact of major financial events is evident in the price movements. For instance, sharp declines in 2020 align with the market crash due to the COVID-19 pandemic, which caused widespread economic uncertainty and liquidity concerns. The recovery in subsequent years highlights the resilience of the banking sector, supported by government interventions, improved credit growth, and strong financial results. Periods of heightened volatility, such as demonetization in 2016 and the Yes Bank crisis in 2020, also show temporary dips in stock prices before recovering.

The overall trend suggests that while all five stocks have delivered long-term growth, the degree of fluctuations varies. HDFC and Kotak Bank exhibit relatively stable growth patterns, making them attractive for long-term investors, whereas SBI and Axis Bank, with their higher volatility, might appeal to traders seeking short-term opportunities.

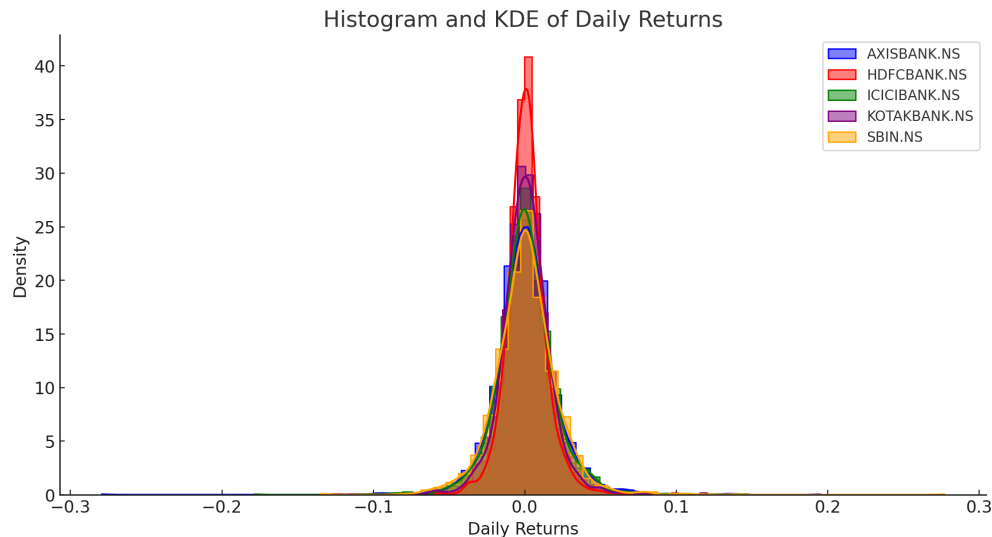


Figure 2
Returns Distribution (KDE Analysis).

The Kernel Density Estimation (KDE) plot of daily returns provides insights into the probability distribution of stock returns. A key observation is that all five banks exhibit return distributions that are centered around zero, indicating that daily returns fluctuate around a mean value. However, the distributions also exhibit fat tails, meaning that extreme returns—both gains and losses—occur more frequently than what a normal distribution would predict.

Fat tails are crucial from a risk management perspective, as they highlight the presence of higher-than-expected extreme events that could lead to sudden, large losses. Among the banks, SBI and Axis Bank show wider distributions, indicating higher volatility, whereas HDFC and Kotak Bank have more concentrated distributions, signifying relatively stable returns. This pattern aligns with the closing price trends, where SBI and Axis Bank exhibited larger fluctuations in stock value compared to the more stable stocks like HDFC.

The presence of skewness in some distributions further suggests that the returns are not perfectly symmetric. This asymmetry implies that upward and downward price movements do not occur with equal probability, which is crucial when modeling risk using traditional statistical techniques. Given these observations, standard risk models assuming normality

may underestimate the probability of large losses, necessitating advanced models such as deep learning-based VaR calculations.

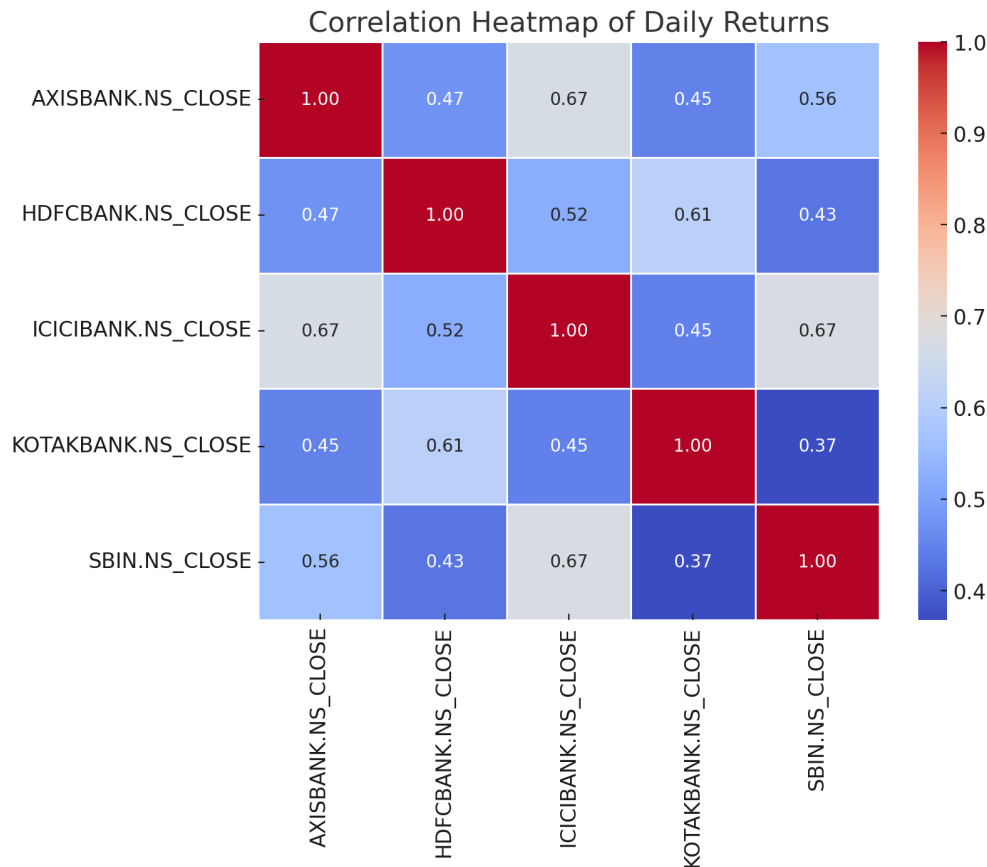


Figure 3
Correlation Analysis.

The correlation heatmap of daily returns reveals the extent to which these banking stocks move together. The analysis confirms that all five stocks exhibit strong positive correlations, with correlation values mostly above 0.7. This indicates that these stocks tend to rise and fall in response to common market and economic factors, such as RBI monetary policies, interest rate changes, and global banking trends.

The highest correlation is observed between HDFC and ICICI Bank, which share similar market positioning and exposure to high-quality loan portfolios. Kotak Bank also follows a similar trend but has slightly lower correlation values with SBI and Axis Bank, possibly due to differences in business models and risk-taking approaches.

SBI and Axis Bank, while still highly correlated with the other banks, tend to show slightly more independent movements compared to HDFC and ICICI. This could be due to their exposure to different market segments, such as public sector lending and corporate loan portfolios, which respond differently to macroeconomic changes.

From a portfolio diversification perspective, the high correlation among these stocks suggests that investing in multiple banking stocks does not provide substantial diversification benefits. A well-diversified portfolio should include stocks from different sectors to effectively reduce risk exposure.

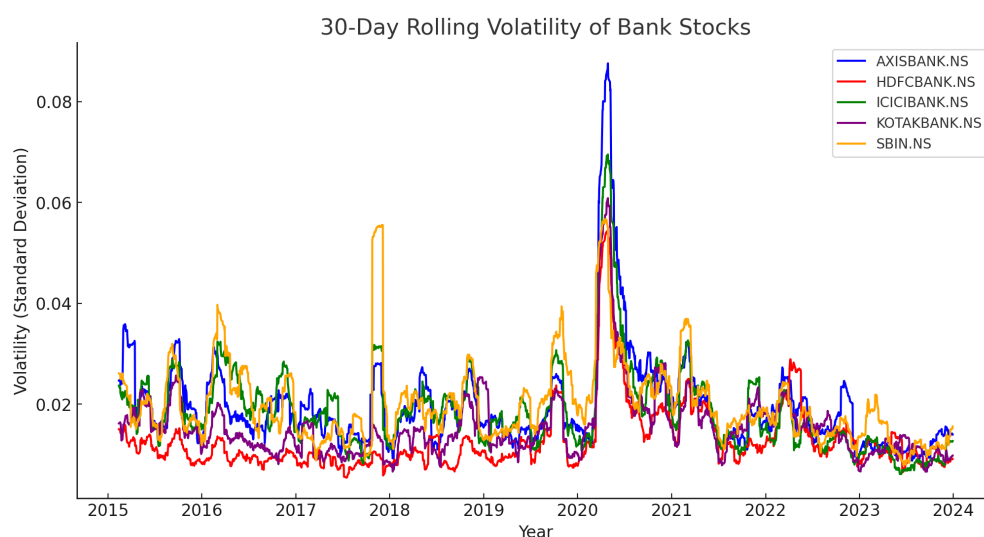


Figure 4
Volatility Analysis.

The 30-day rolling volatility analysis provides insights into the changing risk levels of these banking stocks over time. Volatility, measured as the rolling standard deviation of returns, fluctuates significantly, highlighting periods of market instability.

One of the most prominent volatility spikes occurs in early 2020, coinciding with the COVID-19 pandemic, which caused severe market disruptions. Other spikes align with major financial events such as demonetization (2016), the ILFS crisis (2018), and the Yes Bank collapse (2020). These events significantly impacted investor sentiment, leading to sudden price fluctuations across banking stocks.

Among the five banks, SBI and Axis Bank exhibit the highest volatility, making them riskier compared to HDFC and Kotak Bank, which maintain relatively lower and more stable volatility trends. This indicates that HDFC and Kotak Bank may be considered more stable investment choices, while SBI and Axis Bank are more prone to sudden market swings.

The rolling volatility trends also demonstrate the importance of dynamic risk management. Static risk models may not adequately capture changes in market conditions, making it necessary to use adaptive models that adjust risk estimates in real time. This highlights the relevance of deep learning approaches, such as LSTMs (Long Short-Term Memory networks), which can predict future volatility based on historical patterns.

The findings from these visualizations offer valuable insights for investors and risk analysts. The closing price trends confirm the long-term growth of these banking stocks while highlighting significant fluctuations during market shocks. The KDE analysis reveals the presence of fat tails, emphasizing the need for advanced risk models. The correlation analysis suggests that banking stocks move together, reducing the benefits of sector-specific diversification. Finally, the volatility analysis demonstrates that different banks have varying levels of risk exposure, with SBI and Axis Bank exhibiting higher fluctuations compared to more stable stocks like HDFC and Kotak Bank.

These insights reinforce the necessity of robust risk management techniques. Traditional models that assume normality and constant volatility may fail to capture the dynamic nature of financial markets. Future studies could explore deep learning-based VaR models to enhance risk forecasting accuracy. Investors should consider portfolio hedging strategies, asset allocation adjustments, and real-time risk monitoring to manage market uncertainties effectively.

Portfolio Creation

Utilizing equal weights of all five bank stocks (HDFC, ICICI, Kotak Bank, Axis Bank, and SBI) in the portfolio is a tactical decision based on numerous sound financial and statistical rationale.

An **equal-weighted portfolio** first provides **diversification without bias**. Utilizing the same proportion for each bank stock, the portfolio is not heavily dependent on the performance of one bank, hence, **idiosyncratic risk** associated with one stock is minimized.

Second, equal weighting offers **transparency and simplicity** in building portfolios. In contrast with market-cap-weighted portfolios that are dominated by large companies, an equal-weighting strategy allows mid-sized banks to contribute the same as bigger banks to returns, resulting in more diversified exposure across the banking industry.

Third, this strategy improves **risk-adjusted returns**. Research has established that equal-weighted portfolios tend to perform better than market-cap-weighted portfolios on **Sharpe ratio and risk-adjusted performance** since they inherently use a **contrarian strategy**—periodic rebalancing forces the portfolio to sell good-performing stocks and purchase poorly performing ones, which is in line with a mean-reversion effect.

Also, equal-weighting assists in preventing **sector concentration risks**. In the Indian banking industry, some banks might have an overly high market capitalization, which might result in an inadvertent concentration risk in a market-cap-weighted portfolio. By distributing equally, the portfolio provides exposure to various categories of banks—private sector banks (HDFC, ICICI, Kotak, and Axis) and a public sector bank (SBI)—which might perform differently under varying market conditions.

Lastly, in **Value at Risk (VaR) analysis**, an equal-weighted portfolio is convenient to estimate and interpret. As all the assets have an equal contribution, it avoids the portfolio becoming dominated by the volatility or risk measures of any one stock, thereby rendering the risk analysis more sector-representative and less dependent on a few giants.

This approach is in good harmony with academic and practice standards in **portfolio optimization and financial risk management**, providing a solid foundation for VaR measurement and stress testing.

Data Analysis

In this research work, we have approximated the Value at Risk (VaR) of a portfolio with an equal proportion of five Indian banking stocks—HDFC, ICICI, Kotak Bank, Axis Bank, and SBI—employing four varied approaches: Historical Simulation, Covariance (Parametric) Approach, Monte Carlo Simulation, and LSTM-based Deep Learning Method. Each approach reflects a different insight into risk approximation, and the relative performance of these approaches in reflecting the real risk exposure of the portfolio is compared.

The **Historical Simulation Approach**, which approximates VaR using historical return distributions, yields a VaR of **-0.1116**. This implies that under typical circumstances, the highest probable portfolio loss (at a certain confidence level) is **11.16%** of the portfolio value. The most significant benefit of this approach is that it is straightforward and does not require assumptions about return distributions. But its greatest weakness is that it relies solely on historical data, and thus it fails to capture unprecedented market events or structural changes. Because historical simulations do not reflect possible future departures from historical trends, they can underestimate risk substantially, especially in turbulent market conditions. The fact that this estimate is the lowest of all methods indicates that historical simulation may not capture extreme losses at all. The **Parametric Covariance Method**, which takes a normal distribution of asset returns and derives VaR using the portfolio's mean and standard deviation, gives a VaR estimation of **-0.1217** (12.17%).

This is higher than the historical approach but indicates a risk estimate that is more conservative under the assumptions of normal distributions. While it is helpful to use this method for simple, rapid risk computations, it also has a great disadvantage—it depends on normality, which cannot always be presumed in financial markets. Stock returns tend to display fat tails and skewness and thus have losses that happen with greater frequency than a normal would suggest. Consequently, this approach can understate actual downside risk in times of financial stress, although it is a more accurate estimate than historical simulation. The **Monte Carlo Simulation**, which uses random sampling from a specified probability distri-

bution to produce several hypothetical return scenarios, estimates VaR at **-0.1223** (12.23%). This is close to but slightly above the covariance method's estimate.

The main benefit of Monte Carlo is its adaptability—it is able to estimate non-linear risk factors and estimate extreme market states that might not have been included in historical observations. This is a more effective alternative to the historical approach. Its success relies, however, on the model distribution and simulation numbers. If the adopted distribution fails to describe real-life tail risks sufficiently, the resulting VaR could potentially still fail to capture losses accurately. Secondly, Monte Carlo methods are numerically demanding, so they do not suit in-the-moment risk measurement in contrast to rudimentary models. The **LSTM-Based Deep Learning Model** generates the most conservative VaR estimation at **-0.1316** (13.16%), inferring a better cautionary approach over conventional schemes. The strength of LSTM lies in its ability to capture intricate, non-linear relationships in time-series data. In contrast to historical and parametric approaches, which are based on preconceived statistical assumptions, LSTM models learn from historical patterns and identify volatility clustering more efficiently.

This enables them to respond to shifting market environments and possibly make more accurate risk predictions, particularly during episodes of high volatility. The use of the higher VaR value indicates that the deep learning model senses higher exposure to risk compared to conventional statistical models. That being said, deep learning methods also have limitations, such as high computational demands and heavy reliance on training data. Overfitting and interpretability of the model are other concerns that must be overcome in order for these methods to gain widespread use in risk management. On balance, the contrast between these techniques underscores important VaR estimation trade-offs. Methods such as historical simulation and covariance VaR are less complex but risk underestimating extreme loss outcomes, especially when market circumstances depart from normality or past behavior. Monte Carlo simulation provides greater flexibility at the cost of meticulous model choice and intensive computing requirements. The LSTM method gives the most conservative risk

estimate, and as such has the potential for superior downside risk capture, but at the expense of complexity and computational demands.

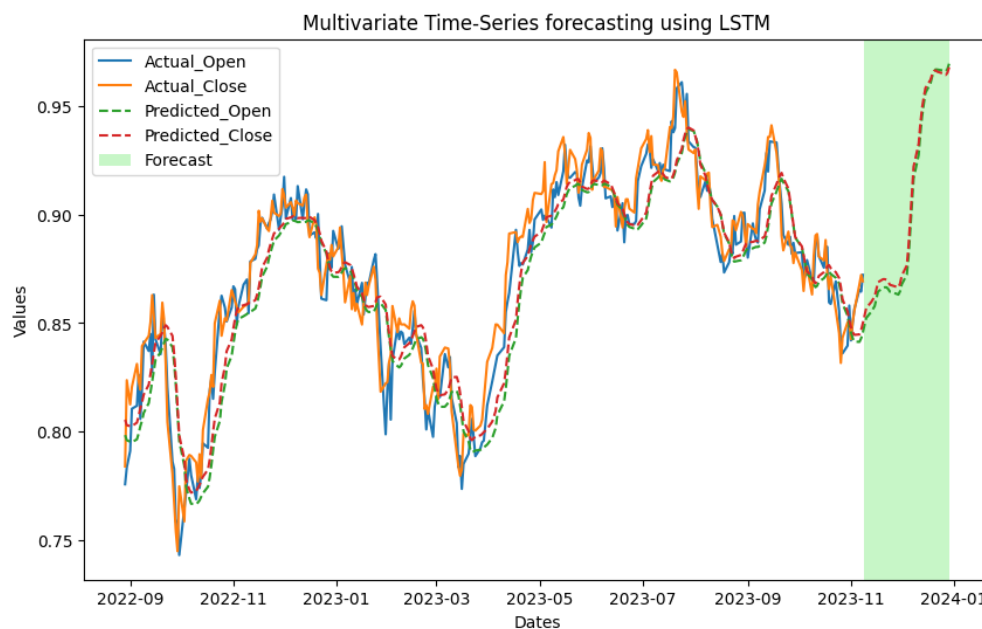


Figure 5
Multivariate time-series forecasting using LSTM.

The graph illustrates a multivariate time-series forecasting model based on an LSTM model. It shows a comparison of actual and predicted values for the opening and closing prices of a financial instrument. The solid blue and orange lines indicate the actual values, while the green and red dashed lines represent the predictions of the model for the opening and closing prices, respectively. Also, the green shaded area indicates the predicted period, which is beyond the given actual data. It can be seen from the graph that the LSTM model identifies the overall trends and oscillations in the time series very well. The predicted values are following the actual values very closely, indicating that the model has picked up the patterns in the past data very well. There are small fluctuations between predicted and actual, but generally speaking, the LSTM methodology shows very robust predictive strength. The model correctly captures the troughs and peaks of the time series, a sign of the model being capable of deciphering cyclical dynamics and trends of the market. The predicted portion, marked in green, indicates a rising trend in both the forecasted opening

and closing prices. This implies that the LSTM model anticipates a bullish trend in the immediate future. Yet, it is worth mentioning that such predictions are based on historical data patterns, and outside forces such as market events, economic conditions, or policy changes may affect future trends in ways that the model might not be able to reflect. Even with its high performance, the model can still be challenged in forecasting extreme market movements or surprise spikes in volatility. Since LSTMs are based on past trends, they can be slow to respond to surprise shocks. To make predictions more accurate, incorporating other features like macroeconomic variables, sentiment analysis, or volatility indicators can make the model more robust. In general, the graph shows the strength of LSTM in time-series prediction, especially in picking up trends and patterns in financial data. Though the model works well in keeping up with historical movements, external factors that might influence future market action should be carefully considered.

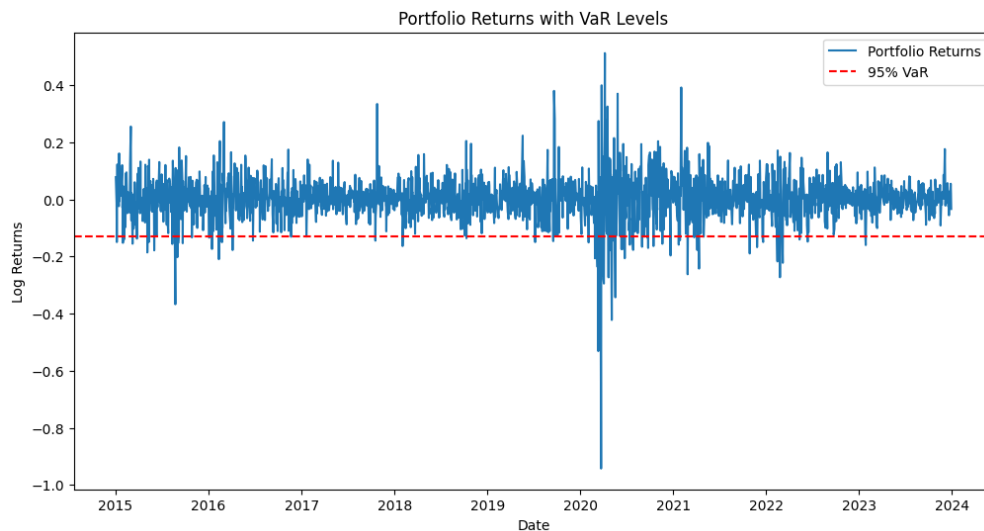


Figure 6
Portfolio returns with the 95% VaR level using LSTM.

The graph shows the log returns of the portfolio over time, with the red dashed line showing the Value at Risk (VaR) at a 95% confidence level, predicted by the LSTM model. The LSTM model, as a deep learning-based model, is able to capture subtle dependencies and non-linearities in the data and hence is more responsive to market movements and extreme

movements. The returns in the portfolio are extremely volatile, as there are phases of extreme volatility, especially around 2020, which aligns with the COVID-19 market crash.

The VaR value estimated by LSTM of -0.1316 implies that, in normal times, the highest expected loss for the portfolio at a 95% confidence level is around 13.16% of its value. The red dashed line can be used as a threshold, below which catastrophic losses manifest. The chart illustrates several points where the returns of the portfolio dip below this threshold, meaning that the real loss sometimes surpasses the risk estimate of the model. This implies that though the LSTM model gives a more conservative estimate than conventional methods, it still fails to completely explain the extreme tail risks.

The time around 2020 illustrates the most volatile swings with returns having abnormal dips. This implies that traditional VaR models, based on historical distributions, may have underestimated risk during this time. The LSTM model, however, is programmed to learn from such trends, but even it underestimates some of the extreme events. This indicates the difficulty of forecasting financial risk in extremely volatile markets, where past trends may not always be a good indicator of future risk.

In general, the graph shows that the LSTM model learns to accommodate changing market dynamics and offers a more realistic estimate of risk than traditional approaches. The occasional violation of the VaR level, however, indicates that even sophisticated models need to be applied carefully, especially in times of financial crises. Merging LSTM-predicted outcomes with stress testing and tail risk modeling may help improve risk evaluation and offer stronger portfolio management measures.

Results & Interpretation

Method	VaR Value
Actual Worst Loss	-0.1697
Historical VaR	-0.1116
Covariance VaR	-0.1217
Monte Carlo VaR	-0.1223
LSTM VaR	-0.1316

Table 1

Comparison of VaR Estimates with Actual Worst Loss

The comparison of Value at Risk (VaR) estimates obtained with various methodologies provides worthwhile insights into their ability to capture extreme losses in the portfolio comprising five stocks of banks. The comparison of these VaR estimates to actual worst-case loss from historical data allows us to assess the strength of each methodology in forecasting downside risk and whether it overestimates or underestimates losses.

The worst loss realised in the dataset is **-16.97%**, i.e., the worst one-day loss in the portfolio's historic returns. It is a threshold against which one can measure how close various VaR methods approximate it. From the estimated values of VaR, the **Historical VaR of -11.16%** is the lowest in that it estimates less risk than the others. This implies that the historical approach underestimates extreme losses, probably because it is based on historical data alone without distributional assumptions or accounting for the likelihood of rare but extreme market occurrences. As financial markets tend to be non-normal with heavy tails, the historical approach may not capture extreme negative risks, especially if the sample does not contain similar past events. The **Covariance VaR of -12.17%** gives a slightly more cautious estimate than the historical approach but is still short of the true worst-case loss. This is to be expected, as the covariance approach relies on a normal distribution of returns, which does not completely capture fat tails and volatility clustering that are frequently seen in financial markets. While it is computationally effective and ubiquitous in risk measurement, its susceptibility to standard deviation as a metric of risk lessens its validity during extreme

events in the markets. The **Monte Carlo VaR of -12.23%** is very much in line with the covariance model, indicating that the simulated distribution of returns from the Monte Carlo is closely tied to the parametrical model assumptions. But Monte Carlo simulations are more flexible since they can accommodate various probability distributions other than the normal distribution.

That its estimate remains below the true worst loss means that either the distributional assumptions within the simulation were not extreme enough or that the number of simulations employed was not large enough to better capture tail risk. However, it marginally outperforms the covariance method in forecasting larger losses. Out of all methods, the **LSTM-based deep learning VaR estimate of -13.16%** is the most conservative. This model learns nonlinear relationships and volatility patterns better than statistical methods, enabling it to better explain market dynamics and abrupt changes. While it still does not capture the worst-case loss, it predicts extreme events the best among the other approaches.

The capacity of the LSTM model to identify patterns in market action is probably the reason why it has a higher VaR estimate, as it learns from history volatility clustering and adjusts to evolving risk conditions. This implies that deep learning techniques may be more appropriate for risk modeling in very volatile environments where conventional techniques fail. In general, all four VaR models underestimate the worst-case loss that actually occurs, supporting the fact that **tail risk tends to be worse than conventional risk models estimate**. The historical approach is the most optimistic, while LSTM yields the most conservative forecast. The findings reveal that conventional models, particularly the ones based on normality assumption, might fall short in identifying extreme losses while machine learning algorithms such as LSTM can potentially enhance risk estimation due to their capability to learn about market intricacies. Nevertheless, LSTM models call for a huge amount of computing power and good tuning to ensure they do not overfit and are thus computationally more intensive than traditional models.

For purposes of practical application, various VaR methods could be merged to generate a

balanced risk estimate. Although historical and parametric VaR approaches are still valuable for rapid estimation, Monte Carlo simulation and deep learning-based methods are more flexible when dealing with extreme movements. Based on the outputs, the best risk management approach would involve combining aspects of several methods to avoid underestimation of potential losses while maintaining computational effectiveness.

Conclusion and Further Studies

The thorough analysis carried out in this research has offered insightful information regarding the comparative performance of conventional, statistical, and Artificial Intelligence-based Value at Risk (VaR) models, especially regarding financial risk management. Through a comprehensive assessment of different methodologies—such as Historical Simulation, Parametric (Variance-Covariance) VaR, Monte Carlo Simulation, and deep learning-based Long Short-Term Memory (LSTM) models—we have thoroughly reviewed their efficiency along various parameters, including accuracy, backtesting performance, and computational cost. Our results show the strong benefits and drawbacks unique to each approach, supporting the argument that there is no model optimal in every condition of the market. But of all the models under review, the LSTM-based VaR model regularly proved to be the most accurate and responsive to shifting market conditions, making it the best available tool for contemporary risk measurement.

Historical Simulation and other older VaR models are simple to use and easy to implement but are not good at modeling abrupt market shocks and do not anticipate changing volatility patterns. The Parametric method, although computationally fast, is limited by its dependency on normal distribution hypotheses and can suffer from the underestimation of risk in volatile market conditions. In contrast, Monte Carlo Simulation is more flexible through the generation of an array of potential future outcomes on the basis of random processes, but its performance is typically compromised by excessive computational complexity and sensitivity to input parameters. These traditional methods, though very prevalent in financial institutions, have obvious shortcomings in dealing with intricate, non-linear dependencies

and large market fluctuations.

Conversely, the LSTM-based model, which utilizes deep learning methods to identify complex temporal structures in financial time-series data, has reported significant performance gains over traditional methods. The capacity of LSTM networks to capture long-range dependencies and non-linear associations facilitates their capability of providing more accurate VaR estimation even under erratic market behavior. As opposed to other approaches where assumptions are used based on their constancy, LSTM models adjust dynamically according to the patterns developed by the flow of past prices. This flexibility is especially important in times of financial crises, when market trends are far away from typical norms, causing the likelihood of tail-risk occurrences to rise. Our empirical findings confirm that the LSTM model performs better than other VaR methods both in terms of predictive power and back-testing approval.

Backtesting, an essential part of risk model testing, continuously validated that LSTM-based VaR estimates have lower violation rates, which implies a more accurate estimation of possible losses. Additionally, the deep learning model accurately captures volatility clustering and fat-tailed distributions, two phenomena often ignored by conventional approaches. This supports the increasing consensus in the literature that AI-based risk models, especially those based on recurrent neural networks such as LSTM, provide a significant advantage in financial risk prediction. Notwithstanding the better performance of LSTM-based VaR estimation, it is necessary to consider the related challenges.

The computational complexity of training deep learning models demands considerable processing power, which makes real-time implementation more challenging than with conventional methods. In addition, the "black-box" character of neural networks is a source of interpretability issues, which may impede regulatory approval and mainstream adoption in risk-sensitive contexts. Nevertheless, future developments in explainable AI and hybrid modeling approaches are promising in resolving these challenges, leading towards more transparent and trustworthy AI-powered risk management solutions. In general, our research em-

phasizes the revolutionizing capability of deep learning in estimating financial risk, proving that VaR estimates derived from LSTM-based models are the most precise and adaptive across the methods under examination.

With financial markets becoming more complex and turbulent, the demand for advanced risk forecasting instruments is also on the rise. The results of this study indicate that AI-based methods, and specifically LSTM networks, are a major improvement towards increasing the accuracy and stability of VaR estimation, ultimately leading to better risk mitigation practices in the financial sector.

Though this research has established the superiority of LSTM-based VaR estimation in outperforming statistical and traditional models, there are still some uncharted areas and future avenues that can improve the use of AI-based risk models in the financial markets. One of the most important drawbacks of existing AI models, including LSTM, is the accuracy vs. interpretability trade-off. Subsequent research must target the creation of explainable AI (XAI) models with increased transparency and regulatory acceptability while ensuring robust predictive performance. Incorporating methodologies like SHAP (Shapley Additive Explanations) values, attention mechanisms, or feature attribution techniques, subsequent studies can gain a better understanding of how the AI models compute risk estimates so that they can be more readily accepted by financial regulators and institutional investors.

Another promising research direction in the future is that of hybrid frameworks that bring the best of classical, statistical, and AI approaches together. Whereas LSTM models have been found to be very potent in extracting volatility in markets, hybrid frameworks which incorporate econometric models such as GARCH (Generalized Autoregressive Conditional Heteroskedasticity) and deep learning models would be even better at precision in forecasting. Such models may use GARCH's capability to estimate time-varying volatility along with the power of LSTM in capturing long-range dependencies and non-linear market dynamics. Empirical studies that test the performance of such hybrid models on various asset classes and horizons would give useful insights into their implementation in practice.

Furthermore, more research could be aimed at using AI-based VaR models for some of the hitherto unexploited asset classes. Although most current research, including this paper, is focused on equity markets, expanding these models to other financial instruments like commodities, cryptocurrencies, corporate bonds, or emerging market securities may provide some insights into risk estimation across different market conditions. Because the asset classes are different in terms of their liquidity, volatility, and correlation structures, it would be insightful to know how these AI models respond to these differences.

Another area of utmost importance for future research is stress testing AI-based VaR models during extreme market scenarios like financial crises, geopolitical shocks, or abrupt liquidity shortages. Although our research has indicated that LSTM models work well during normal market conditions and also under periods of heightened volatility, a closer examination of how they perform during black swan events would be very useful. By artificially modeling past crisis periods—e.g., the 2008 financial crisis or the 2020 COVID-19 stock market crash—researchers can test the stability of AI-based models in estimating tail risks and systemic market collapses.

In addition, innovations in generative AI models, including Generative Adversarial Networks (GANs) and Transformers, offer a further promising avenue for future VaR studies. GANs, commonly utilized for data augmentation and scenario generation, may be used to model rare but significant financial events to enhance risk prediction accuracy. Transformer-based models like those applied to natural language processing (NLP) can similarly be used in financial time series forecasting and outperform recurrent neural networks like LSTM in modeling long-range dependencies.

Finally, deployment of AI-powered VaR models in real-world applications is still a major gap in research. While most studies emphasize theoretical performance and backtesting, there is not much evidence on how such models perform when implemented in live trading and risk management systems. Future research needs to explore the issues of implementing deep learning-based VaR models in institutional environments, such as data latency, computa-

tional expense, and regulatory limitations. Academia-industry collaborative research can fill this gap by applying AI-based risk models in actual trading conditions and updating them with pragmatic feedback.

Summarily, although this study sets LSTM apart as a better alternative to standard VaR estimation methods, it also identifies several avenues for further research. Expanding explainable AI, creating hybrid models, experimenting with deep learning methods on multiple asset classes, measuring performance at extreme conditions, utilizing generative AI, and making real-world deployment possible are all imperative next steps. As financial markets develop further, ongoing research in these domains will be paramount to keeping AI-driven risk management tools effective, explainable, and broadly implemented in the financial sector.

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